StructuraMiner

Exhaustive Structural Feature Extraction from Protein Data Bank Files

Version 1.0.0 | User Manual & Technical Reference

StructuraMiner is a production-grade, multi-level computational framework that exhaustively mines structural, physicochemical, topological, and interaction-based features from any Protein Data Bank (PDB) file. Designed for structural bioinformatics, machine learning pipelines, and comparative proteomics, it transforms raw atomic coordinates into rich, analysis-ready tabular datasets across seven complementary levels of protein organisation.

7 Output Modules	~120+ Features/Residue	ML-Ready CSVs
Auto-Installs Deps	**Multi-Core Parallel**	**Full & Stripped PDB**

pdb_feat.py | *MIT License* | *Python 3.8+* | *github.com/<your-username>/StructuraMiner*

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1. Overview and Motivation

Protein structure encodes function. Yet extracting machine-readable, multi-scale structural features from raw PDB files is a laborious, error-prone task that typically requires integrating half a dozen specialised tools. **StructuraMiner** solves this by providing a single, self-contained Python script that exhaustively characterises a protein structure across seven distinct levels of organisation — from individual atoms to the whole molecule — and serialises every result into tidy, analysis-ready CSV files.

The framework is purposefully designed to be dependency-free at invocation: missing packages are detected and installed automatically before any computation begins. It accepts both full RCSB-format PDB files (with HEADER, REMARK, SEQRES, HELIX, SHEET, SSBOND records) and stripped ATOM-only files, and harmonises both into a uniform internal representation via an intelligent format detector.

Primary use cases include: structural bioinformatics research, protein engineering, ML/deep-learning feature pipelines, comparative proteomics, drug-target characterisation, and quality assessment of predicted structures (e.g. AlphaFold models).

2. Installation and Dependencies

StructuraMiner requires Python 3.8 or later. All Python dependencies are auto-installed on first run; however, pre-installing them is recommended for reproducibility:

```
pip install biopython numpy pandas scipy networkx freesasa pydssp tqdm
```

Package	Version	Role in StructuraMiner
biopython	≥ 1.79	PDB parsing, dihedral/angle calculation, NeighborSearch
numpy	≥ 1.21	Coordinate arrays, PCA, distance computations
pandas	≥ 1.3	Tabular output, CSV serialisation
scipy	≥ 1.7	ConvexHull, KDTree, pairwise distances, statistics
networkx	≥ 2.6	Protein contact network, centrality metrics
freesasa	≥ 2.1	Solvent-accessible surface area (SASA) calculation
pydssp	≥ 0.8	DSSP-style secondary structure assignment
tqdm	≥ 4.60	Progress bars for long-running steps

3. Command-Line Usage

StructuraMiner is invoked from the command line as follows:

```
# Basic run (single core)
python pdb_feat.py --input protein.pdb
```

```
# Multi-core run with custom output prefix
python pdb_feat.py --input protein.pdb --cores 8 --output my_run

# Suppress progress output
python pdb_feat.py --input protein.pdb --cores 4 --quiet
```

Argument	Short	Default	Description
--input	-i	required	Path to input PDB file
--cores	-c	1	CPU cores for parallel extraction
--output	-o	PDB stem	Prefix for all output file names
--quiet	-q	False	Suppress verbose timestamped logging
--version			Print version number and exit

4. Framework Architecture and Pipeline Stages

StructuraMiner is organised as eight sequential pipeline stages, each encapsulated in its own Python class. The master orchestrator **PDBFeatureFramework** coordinates all stages and handles I/O.

Stage 1: PDBLoader

Detects PDB format (full RCSB or ATOM-only), loads the structure via BioPython PDBParser, and parses all raw text records (HEADER, SEQRES, HELIX, SHEET, SSBOND, SITE, CRYST1). Exposes a unified model object to all downstream extractors.

Stage 2: PerResidueExtractor

The most feature-rich stage. Computes ~120 features per amino-acid residue by running: FreeSASA for solvent exposure, pyDSSP for secondary structure assignment, BioPython PPBuilder for backbone dihedral angles (phi/psi/omega/chi1/chi2), inter-residue contact analysis (CA, CB, heavy-atom), hydrogen bond detection, salt bridge identification, disulfide bond detection, pi-pi stacking, cation-pi interactions, backbone bond geometry, B-factor statistics, NetworkX protein contact network centrality metrics, water contacts, and neighbour composition.

Stage 3: GlobalFeatureExtractor

Aggregates protein-wide descriptors: sequence composition and class fractions, molecular weight, net charge and pI (Henderson-Hasselbalch titration), mean hydrophobicity (Kyte-Doolittle and Eisenberg scales), geometric descriptors (radius of gyration, convex hull volume/area, PCA eigenvalues, asphericity, acylindricity, relative shape anisotropy, prolateness), B-factor distribution statistics, and secondary structure content fractions.

Stage 4: PerAtomExtractor

Records element identity, van der Waals radius, mass, partial B-factor, 3D coordinates, and residue membership for every heavy atom in the structure.

Stage 5: ChainFeatureExtractor

Aggregates per-residue features at the chain level: length, residue composition, mean SASA, mean B-factor, secondary structure content, charge, and hydrophobicity per chain.

Stage 6: InteractionFeatureExtractor

Enumerates all pairwise non-covalent interactions: hydrogen bonds (N-O distance < 3.5 Å), disulfide bonds (SG-SG < 2.2 Å), salt bridges (charged atoms < 4.0 Å), pi-pi stacking (centroid < 7 Å, ring-plane angle < 30°), cation-pi contacts (within 6 Å of ring centroid), and hydrophobic contacts (CA-CA < 5 Å between hydrophobic residues). Each row carries interaction type, partner identities, and distance.

Stage 7: SecondaryStructureDetailExtractor

Parses declared HELIX and SHEET records from PDB REMARK lines and computes per-segment statistics including residue count, helix type, strand sense, and mean B-factor.

Stage 8: DistanceMatrixExtractor

Computes the complete upper-triangle CA-CA pairwise distance matrix and exports it in long format with sequence separation. Essential for contact-map and distance-map analyses.

5. Output Files — Structure and Content

StructuraMiner produces eight output files, all named after the input PDB stem (or the user-supplied --output prefix). Each file is described below with its column structure and biological interpretation.

5.1 Global Features ({stem}_global_features.csv)

A single-row CSV capturing the protein's holistic properties. This file is the first place to look for an overall characterisation of the structure.

Column Group	Key Columns	Biological Meaning
Header / metadata	pdb_id, title, resolution_A, structure_method, space_group	Provenance, experimental resolution, crystal symmetry
Composition	n_residues, count_*/frac_* (per AA), n_chains, n_water, n_hetatm	Sequence length, amino-acid usage, ligand/water content
Physicochemistry	mw_da, net_charge_pH7, pI_approx, frac_hydrophobic, mean_hydrophobicity_KD/Eisenberg	Molecular weight, charge, solubility prediction
Geometry / Shape	radius_of_gyration_A, end_to_end_dist_A, convex_hull_volume_A3, pca_eigenval_1-3, asphericity, relative_shape	Compactness, elongation, globularity
B-factor statistics	bfactor_CA_mean/std/skew, bfactor_all_heavy_IQR	Global thermal motion, crystallographic quality
Secondary structure	ss_frac_helix/sheet/coil, ss_n_helix/sheet_segments	Structural class (alpha/beta/mixed), fold topology
Crystallography	unit_cell_a/b/c, unit_cell_alpha/beta/gamma	Crystal packing parameters

5.2 Per-Residue Features ({stem}_per_residue_features.csv)

The core output: one row per amino-acid residue, approximately 120 columns. This file enables residue-level ML models, hotspot prediction, Ramachandran analysis, and buried/exposed classification.

Feature Group	Columns	Insight Provided
Identity	chain, resnum, resname, one_letter, residue_class	Residue location and classification
SASA (FreeSASA)	sasa_total, sasa_backbone, sasa_sidechain, sasa_relative, is_buried, is_surface	Solvent exposure — buried core vs. surface residues
Secondary Structure	ss_code (H/E/-)	DSSP-style assignment for each residue
Backbone dihedrals	phi_deg, psi_deg, omega_deg	Ramachandran plot coordinates; cis-peptide detection
Side-chain dihedrals	chi1_deg, chi2_deg	Rotamer state; side-chain packing quality
Contacts (CA, 8 Å)	n_contacts_CA, n_short/medium/long_contacts, mean_contact_dist	Local packing density and sequence context
Contacts (heavy, 5 Å)	n_contacts_heavy5	Tight atomic packing
Hydrophobic contacts	n_hydrophobic_contacts	Hydrophobic core participation
Hydrogen bonds	n_hbond_donor, n_hbond_acceptor, n_hbond_total	H-bond network contribution
Salt bridges	n_salt_bridges	Electrostatic stabilisation
Disulfide bonds	in_disulfide, ss_partner_chain, ss_partner_resnum	Covalent cross-links

Pi-pi stacking	n_pipi_stacking	Aromatic ring interactions
Cation-pi	n_cation_pi	Electrostatic aromatic interactions
Backbone geometry	bond_CA_N, bond_CA_C, bond_C_O, angle_N_CA_C, angle_CA_C_O, ca_displacement	Structural distortion; outlier detection
B-factors	bfactor_res_mean/max/std, bfactor_backbone, bfactor_sidechain	Residue-level thermal motion; flexible regions
Network centrality	network_degree, network_betweenness/closeness/eigenvector_centrality, network_clustering_coeff	Protein contact network, hubs, allosteric hotspots
Water contacts	n_water_contacts, min_water_dist_A	Hydration shell; buried water sites
Neighbour composition	nbr_frac_aliphatic/aromatic/polar/charged...	Microenvironment of each residue
Physicochemical	hydrophobicity_KD/Eisenberg, vdw_volume, residue_formal_charge	Residue-level biophysical properties

5.3 Per-Atom Features ({stem}_per_atom_features.csv)

One row per heavy atom. Columns include: chain, residue number and name, atom name, element symbol, van der Waals radius, estimated mass, B-factor, occupancy, and 3D coordinates (x, y, z). This file is the basis for atomic-resolution analyses such as packing calculations, void detection, and graph neural network featurisation where nodes represent individual atoms.

5.4 Per-Chain Features ({stem}_per_chain_features.csv)

One row per polypeptide chain. Aggregates from per-residue data: chain length, amino-acid composition, mean and total SASA, mean B-factor, secondary-structure content fractions, net charge, and mean hydrophobicity. Essential for multi-chain complex analysis, interface characterisation, and chain-level comparisons in oligomeric proteins.

5.5 Interactions ({stem}_interactions.csv)

Each row describes one pairwise non-covalent interaction. Columns: **interaction_type** (hydrogen_bond, disulfide, salt_bridge, pi_pi_stacking, cation_pi, hydrophobic_contact), chain/resnum/resname for both partners, specific atom names, and measured distance in Å. This file directly feeds protein interaction network analyses, binding-site characterisation, and stability engineering studies.

5.6 Secondary Structure Segments ({stem}_ss_segments.csv)

One row per declared secondary-structure element (HELIX or SHEET record). Columns: element_type, helix_id/sheet_id, chain, start, end, length (number of residues), helix_type (1=alpha, 3=Pi, 5=3/10), strand sense (parallel/antiparallel), number of residues confirmed in structure, and mean B-factor of the segment. Enables topology-level comparisons, helix-length distributions, and segment-quality assessment.

5.7 CA Distance Matrix ({stem}_distance_matrix.csv)

Long-format upper-triangle pairwise CA-CA distance matrix. Columns: chain1, resnum1, resname1, chain2, resnum2, resname2, ca_dist_A, seq_separation. The seq_separation column (|i – j|) enables easy filtering of short-range vs. long-range contacts. This file is the direct input for contact-map visualisation, distance-map ML models, and topological domain analysis.

5.8 Extraction Summary Report ({stem}_extraction_summary.txt)

A human-readable plain-text summary printed at the end of every run. Captures: PDB metadata (ID, title, resolution, method), composition (chain/residue/atom/water counts), shape (radius of gyration, MW, pI), secondary structure fractions, a manifest of all output file paths, total extraction time, and a timestamp. Serves as a provenance record for reproducibility.

6. Biological Significance of Key Features

Solvent-Accessible Surface Area (SASA)

Computed by FreeSASA using the Lee-Richards rolling-probe algorithm. Relative SASA (rSASA) classifies each residue as buried (rSASA < 0.25), intermediate, or surface-exposed (rSASA > 0.40). Buried residues form the hydrophobic core critical for stability; surface residues drive interaction specificity and are prime targets for mutagenesis.

Backbone Dihedral Angles (phi, psi, omega)

Phi and psi define Ramachandran space; allowed regions correspond to alpha-helices (-60°, -45°), beta-strands (-120°, 120°), and left-handed helices. Outliers indicate model errors or genuine conformational strain. Omega (near 180°) detects rare cis-peptide bonds, which have functional relevance in enzyme active sites.

Protein Contact Network Centrality

StructuraMiner builds a CB-CB contact graph (< 8 Å) and computes five centrality metrics. Residues with high betweenness or eigenvector centrality are structural and/or allosteric hubs: mutating them disproportionately perturbs folding or signal transmission. This analysis has direct applications in rational drug design and directed-evolution campaigns.

B-factors (Crystallographic Temperature Factors)

B-factors report atomic displacement from the mean position. High B-factors indicate flexible, disordered regions; low values mark the rigid core. StructuraMiner extracts per-residue B-factor statistics (mean, max, std) separately for backbone and side-chain atoms, enabling fine-grained flexibility profiling and comparison across homologues.

Non-Covalent Interactions (H-bonds, Salt Bridges, Pi stacking)

The interactions CSV enumerates every detected non-covalent contact. Hydrogen bonds stabilise secondary structure and enzyme active sites. Salt bridges contribute 1–5 kcal/mol per pair to thermostability. Pi-pi stacking (parallel aromatic rings < 7 Å) and cation-pi interactions (within 6 Å) are major contributors to protein-ligand and protein-protein binding affinity.

Isoelectric Point (pI) Estimation

Computed via 150-iteration Henderson-Hasselbalch bisection, accounting for all ionisable side chains (Asp, Glu, His, Cys, Tyr, Lys, Arg) and terminal charges. pI governs solubility, crystallisability, and interaction propensity at physiological pH — critical for biopharmaceutical development and purification strategy.

7. Use Cases and Applications

7.1 Machine Learning Feature Engineering

The per-residue CSV provides a ready-made feature matrix for residue-level classifiers (solvent exposure, secondary structure, binding-site prediction). The global CSV serves as a molecular fingerprint for protein-level regressors (e.g. melting temperature, solubility, or activity prediction).

7.2 Structural Quality Assessment

B-factor distributions, Ramachandran outlier rates, and backbone bond-length deviations can be used to benchmark experimental structures or computationally predicted models (AlphaFold, RoseTTAFold) against high-resolution references.

7.3 Comparative Proteomics

Running StructuraMiner across a family of homologous structures generates aligned feature tables suitable for evolutionary analysis: which residues are conserved in the buried core, how does secondary structure content vary with thermostability, etc.

7.4 Drug Discovery and Binding-Site Mapping

The SITE records, interaction network, and SASA values combinedly pinpoint druggable pockets: surface-exposed cavities with local hydrophobicity, enriched in aromatic or charged residues, and served by high-centrality hub residues.

7.5 Protein Engineering and Stability Optimisation

Salt-bridge counts, disulfide-bond mapping, packing-ratio, and hydrophobic-contact density guide rational engineering campaigns. The distance matrix enables loop-modelling and linker-design decisions.

8. Limitations and Notes

StructuraMiner processes model 0 (the first MODEL block) of multi-model PDB files (e.g. NMR ensembles). To analyse all NMR models, split the file beforehand and run the tool on each.

Hydrogen-bond detection uses a simplified distance criterion (donor-acceptor ≤ 3.5 Å) rather than full H-atom placement. For high-precision H-bond enumeration, preprocess the structure with Reduce or MolProbity.

The pI calculation is an approximation using standard solution-phase pKa values. Microenvironment shifts (buried titratable residues, metal coordination sites) are not modelled.

The CA distance matrix has $O(N^2)$ size. For large structures (> 2,000 residues) this file can be substantial in disk space. Consider filtering by seq_separation or distance threshold for downstream use.

FreeSASA and pyDSSP both require well-formatted ATOM records. Heavily non-standard or truncated PDB files may cause these sub-modules to fall back gracefully to None values, which are flagged in the CSV.